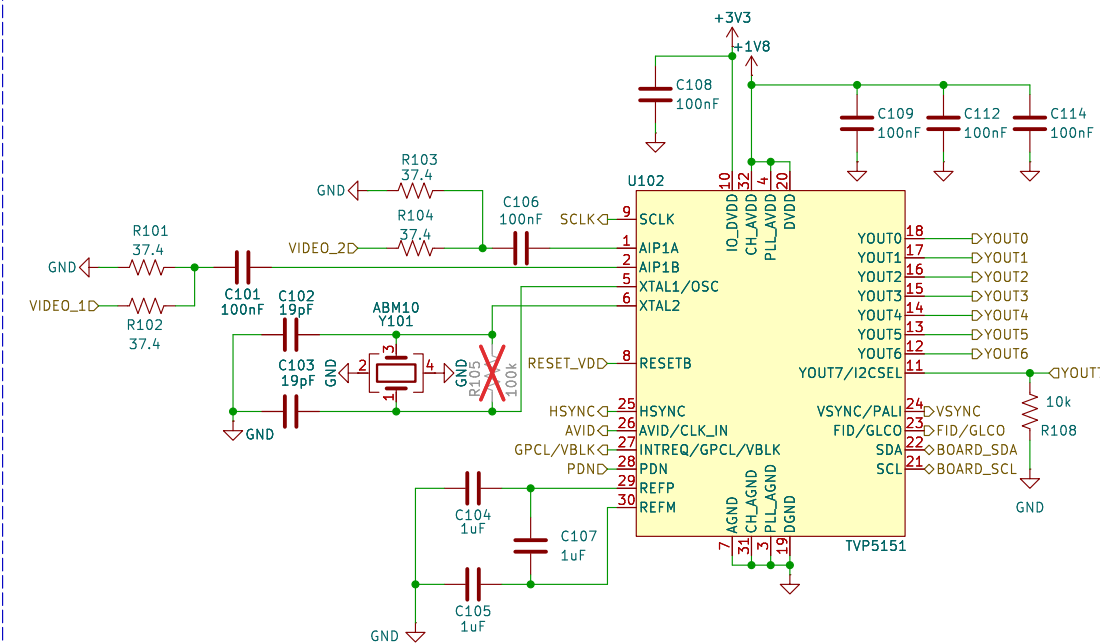


Video Decoder

TVP5151PBS

I2C Address:
1011100 (0x5C)

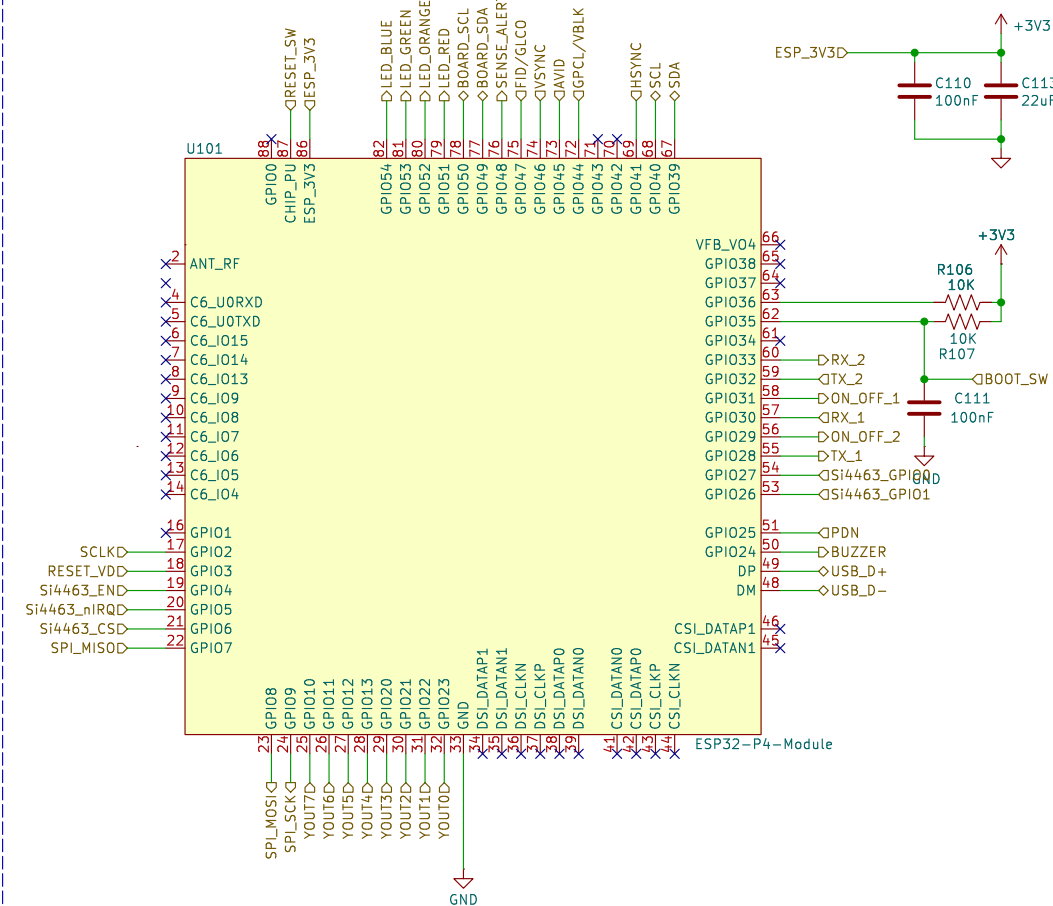


Crystal

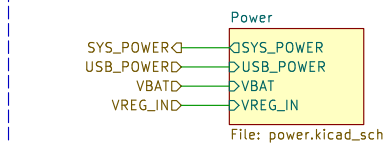
ABM10-27.000MHZ

Microcontroller Module

ESP32-P4



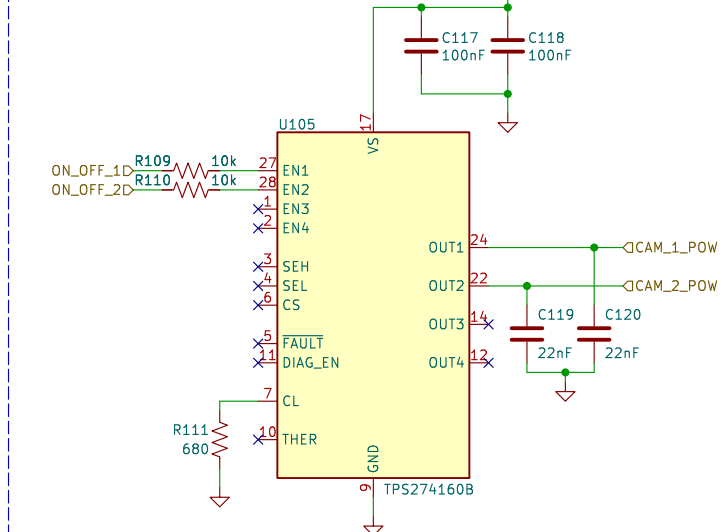
Hierarchical Power Circuitry



(Global 3V3, 5V5, 1V8 flag power outputs inside)

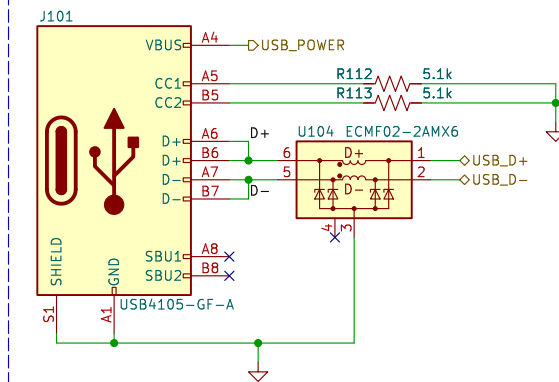
Camera Toggle

TPS274160B



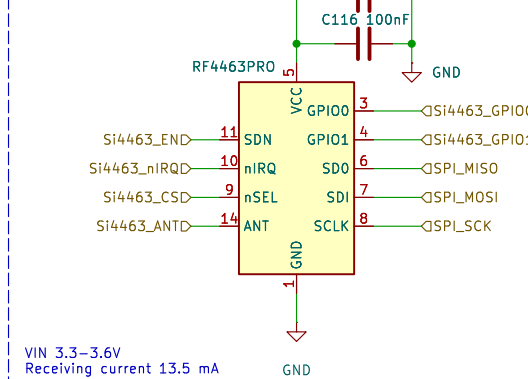
Current Limit: 2.94A

USB-C 2.0 Connector



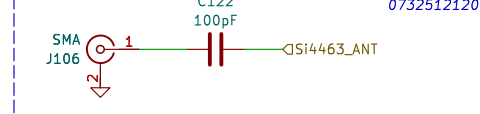
Transceiver Module

RF4463PRO

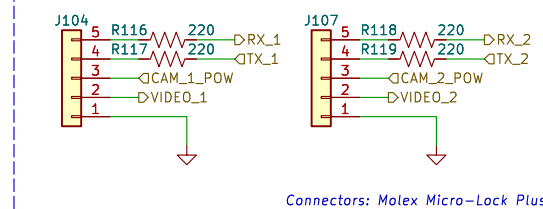


VIN 3.3-3.6V
Receiving current 13.5 mA

Antenna

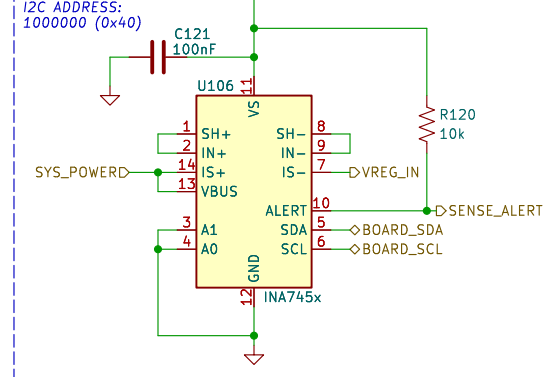


Camera/Transceiver Connectors

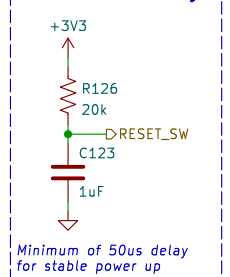


BATTERY SENSE

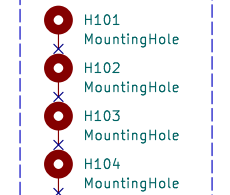
INA745B



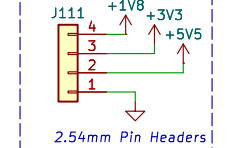
MCU Boot Delay



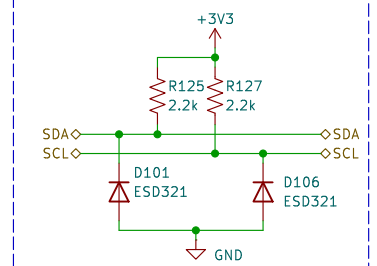
Mounting Holes



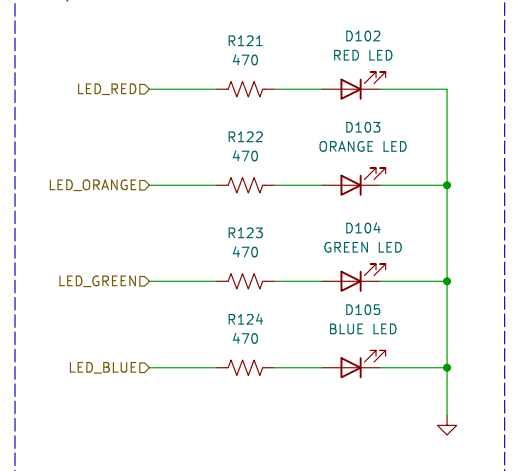
Power Debug Pins



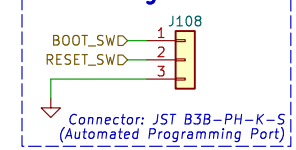
I2C Pull-Up Resistors (for Midas)



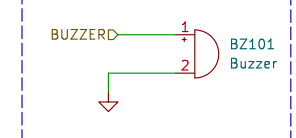
STATUS LEDs



MCU Debug Pin



PIEZO BUZZER



Contributors: Eddie Tang, Rohan Das, Thomas McManamen, Chethan Karandikar, Andrew Dorman, Mason Miao, Kacper Paraniuk, Azalea Shillington, Mike Hung, Sahir Pagey

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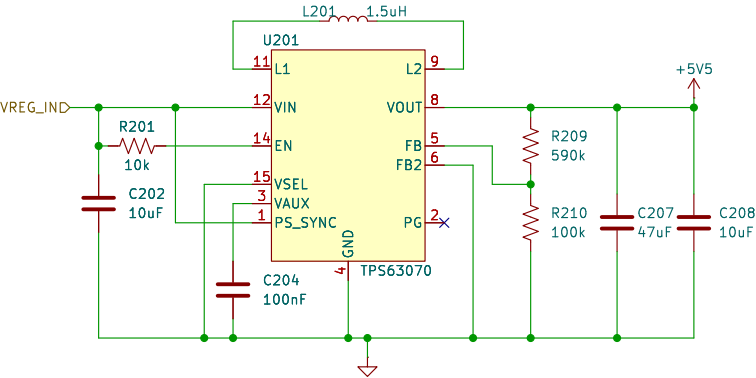
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Title: CAM MK3 Schematics

Size: A3
Date: 2025-12-02
KiCad E.D.A. 9.0.6

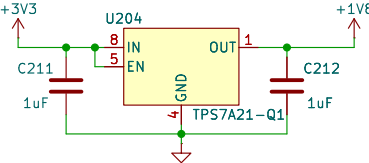
Rev: A
Id: 1/2

5V5 Voltage Regulator
TPS63070



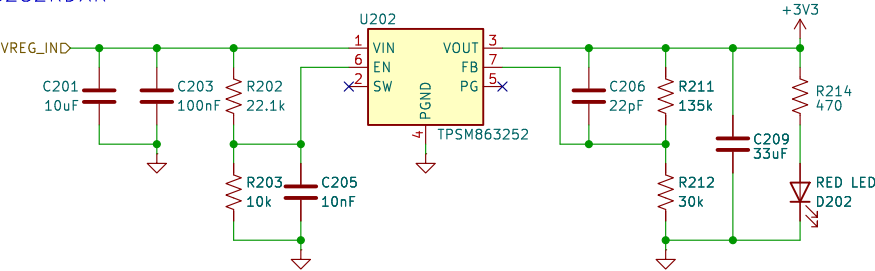
VIN 5 - 8.4V
5.5V VOUT
2A Max Current

1.8V Voltage Regulator
TPS7A21-Q1



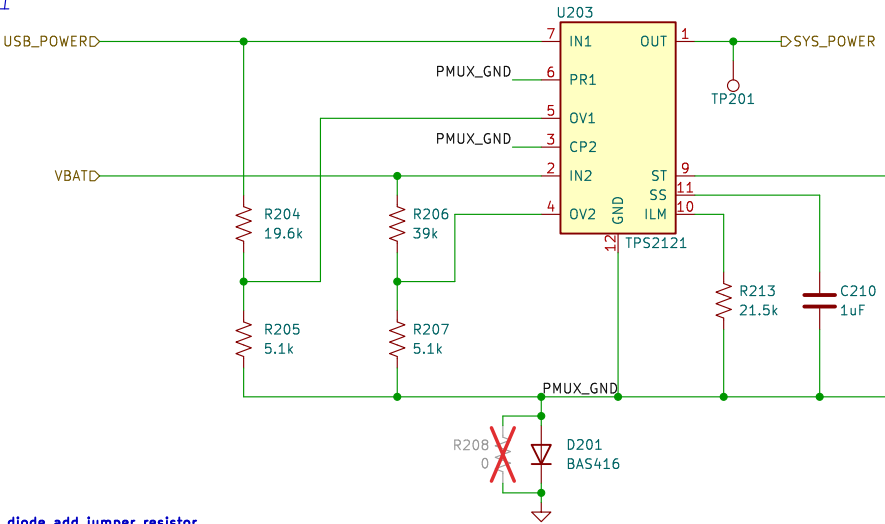
VIN 2 - 6V
1.8V VOUT
Max IOut 0.5A

3V3 Voltage Regulator
TPSM863252RDXR



VIN 5-8.4V
3V3 VOUT
Max IOut 3A

Power MUX
TPS2121



USB_Power OV: -5.54V
VBAT OV: -10.2V
OC: -4.5A

If not using diode add jumper resistor
and recalculate resistor voltage dividers

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Sheet: /Power/
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Title: CAM MK3 Power Schematics

Size: A3 Date: 2025-12-02

KiCad E.D.A. 9.0.6

Rev: A

Id: 2/2